STA 198/GLHLTH 198 - Introduction to Global Health Data Science

Fall 2026

Table of contents

Short Summary **2**

Course Overview . . . 2
Required Materials . . . 2
Assessment & Grading . . . 2
Key Policies . . . 3
Academic Honesty . . . 3
Deadlines & Late Work . . . 3
Attendance . . . 3
Important Dates . . . 3
Getting Help . . . 3

Details **4**

Textbooks . . . 4
Computing . . . 4
Accessibility . . . 4
Communication . . . 4
Email . . . 5
Tips for success . . . 5
Getting help . . . 6
Course components . . . 6
Lectures . . . 6
Labs . . . 7
Teams . . . 7
Activities & Assessment . . . 7
Assessment & Grading . . . 8
Course policies . . . 9
Duke Community Standard . . . 9

Academic honesty 9
Regrade requests 11
Accommodations 11
Academic accommodations 11
Religious accommodations 11

Short Summary

Course Overview

Learn to explore, visualize, and analyze data using R and RStudio through real-world global health problems. Study best practices in data wrangling, visualization, modeling, and statistical inference. Communicate results in plan language accessible to non-experts. Distinguish good evidence from weak or misleading evidence using statistical reasoning.

Required Materials

- **Textbooks** (free online): *R for Data Science, 2e* and *Introduction to Modern Statistics*; (free in library reserves and optional): *Principles of Biostatistics*
- **Computing**: Laptop required for all classes; R accessed through Duke containers, no installation needed
- **Platform**: Course website at sta198-f26-2.github.io – everything you need is there or linked from there!

Assessment & Grading

Component	Weight	Details
Lectures	5%	Attendance/participation (20 non-exam lectures minimum for full score)
Labs	20%	Individual or team-based, completed in class (lowest 2 dropped)
Exam 1	25%	In-class
Exam 2	25%	In-class
Exam 3	25%	In-class
Final	Replace lowest exam if desired	In-class

Key Policies

Academic Honesty

- **Individual work**: Exams must be completed alone
- **Team collaboration**: Expected for some labs
- **AI tools**: Allowed for code assistance with proper citation
- **Online resources**: Permitted with explicit citation

Deadlines & Late Work

- **Labs**: No late work (completed in class)
- **Exams**: No make-ups (final can substitute for lowest score)

Attendance

- **Lectures**: Mandatory attendance
- **Labs**: Mandatory attendance

Important Dates

- **Classes begin**: Aug 24
- **Exam 1**: Sept 30
- **Exam 2**: Nov 4
- **Exam 3**: Dec 2
- **Classes end**: Dec 4
- **Final exam**: Dec 11

Getting Help

- **Office hours**: Regular TA and instructor availability
- **Ed Discussion**: Online forum for course questions
- **Email**: Include "STA 198" in subject line
- **Accommodations**: Contact SDAO for academic accommodations

Details

Textbooks

The first two books are **freely available online**.

- R for Data Science, 2e, Wickham, Çetinkaya-Rundel, Grolemund. O'Reilly, 2nd edition, 2023. Hard copy available on Amazon.
- Introduction to Modern Statistics, Çetinkaya-Rundel, Hardin. OpenIntro Inc., 2nd Edition, 2023. Hard copy available on Amazon.

The last book is not required. Three copies are on reserve at the library, and the problem sets and "further applications" worked examples in each chapter are *quite* useful for students wanting more examples and problems (I suggest everyone check out the further applications!).

- Principles of Biostatistics, Pagano, Gauvreau, and Mattie. Taylor and Francis, 3rd Edition, 2022.

Computing

You will need a laptop you can bring to lecture and lab for this course. We will use the statistical software R. Students will be able to access R through Docker containers provided by Duke Office of Information Technology. See the computing page for more information.

Accessibility

If any portion of the course is not accessible to you due to challenges with technology or the course format, please let me know so we can make appropriate accommodations.

The Student Disability Access Office (SDAO) is available to ensure that students can engage with their courses and related assignments. Students should contact the SDAO to request or update accommodations under these circumstances. If you have an exam accommodation, please schedule a seating there for the three exams and final as soon as possible.

Communication

All lecture notes, assignment instructions, an up-to-date schedule, and other course materials may be found on the course website: sta198-f26-2.github.io.

Announcements will periodically be emailed through Canvas Announcements. Please check your email regularly to ensure you have the latest announcements for the course.

Email

If you have questions about assignment extensions, accommodations, or any other matter not appropriate for the class discussion forum, please email me directly at ahh24@duke.edu. **If you do so, please include "STA 198" in the subject line.** Barring extenuating circumstances, I will respond to STA 198 emails within 48 hours, Monday through Friday. Response time may be slower for emails sent Friday evening through Sunday.

Tips for success

Your success in this course depends very much on you and the effort you put into it. Your TA and I will help you by providing you with materials answering questions, and setting a pace, but for this to work, you must do the following:

1. **Keep up with the material.**

Come prepared by doing readings so you can deeply engage with the material during lectures and labs. The schedule lists extensive additional resources if you need additional support – take advantage if needed so you don't fall behind.

2. **Be present and engaged in every lecture and lab.**

In lectures ask questions and participate in discussions. In labs, work on the lab exercises, ask questions, and collaborate with your teammates – this is your chance for coding with strong support from your peers and the TA. If you miss a class, make sure to catch up on the material before the next class.

3. **Ask questions.**

As often as you can. In class, out of class. Ask me, ask the TAs, ask your friends, ask the person sitting next to you. This will help you more than anything else. If you get a question wrong on an assessment, ask us why. If you're not sure about the lab, ask. If you hear something on the news that sounds related to what we discussed, ask. If the reading is confusing, ask. If something seems overwhelming, ask.

4. **Do the work.**

The earlier you start, the better. It's not enough to just mechanically plow through material. You should ask yourself how things relate to earlier material and imagine how they might be changed (to make questions for an exam, for example).

5. **Don't procrastinate.**

The content builds upon what was taught in previous weeks, so if something is confusing to you in Week 2, Week 3 will become more confusing, Week 4 even worse, etc. Don't let the week end with unanswered questions. But if you find yourself falling behind and not knowing where to begin asking, come to office hours and work with a member of the teaching team to help you identify a good (re)starting point.

Getting help

- If you have a question during the lecture or lab, feel free to ask it! There are likely other students with the same question, so by asking, you will create a learning opportunity for everyone.
- The teaching team is here to help you be successful in the course. You are encouraged to attend office hours to ask questions about the course content and assignments. Many questions are most effectively answered as you discuss them with others, so office hours are a valuable resource. Please use them!
- Outside of class and office hours, any general questions about course content or assignments should be posted on the class discussion forum, Ed Discussion. There is a chance another student has already asked a similar question, so please check the other posts on the forum before adding a new question. If you know the answer to a question, I encourage you to respond!

Check out the Support tab for more resources.

Course components

Lectures

Lectures designed to be interactive, so you gain experience applying new concepts and learning from each other. My role as instructor is to introduce you to new methods, tools, and techniques, but it is up to you to take them and use them. A lot of what you do in this course will involve writing code, and coding is a skill that is best learned by doing. Therefore, many lectures will feature application exercises that will serve as opportunities to practice what you're learning as you're learning it and be great preparation for the assignments and exams.

You are expected to bring a laptop (or Chromebook) to each class so that you can participate in the application activities. Please ensure your device is fully charged before you come to class, as the number of outlets in the classroom will not be sufficient to accommodate everyone. A tablet also works, but the user experience will be much smoother on a laptop.

Labs

Labs are designed to be hands-on at all times, therefore you're similarly expected to bring a laptop to each lab session.

During labs you will get a brief introduction to the week's assignments from your TA and then work on the lab exercise, either individually or with your teammates.

Teams

Many labs will involve teamwork. All team members are expected to contribute equally to the completion of the lab, and you will be asked to evaluate your team members throughout the semester. Failure to adequately contribute to any component will result in a penalty to your mark relative to the team's overall mark. You are expected to use the provided GitHub repository as the central collaborative platform. Commits to this repository will be used as a metric (one of several) of each team member's relative contribution to each lab.

Activities & Assessment

Lectures

Attendance and participation will be monitored throughout the semester. Your attendance and participation score will be calculated as the percentage of lectures in which you engaged in person (i.e., sleeping or online shopping does not count, and you must participate in class activities). The denominator used for this calculation will be 22 (we have 22 non-exam lecture days, excluding the first day of class). Students who are active in 20 or more lectures will receive full credit. That is, if you actively attend all 22 lectures after the first day of class and the constellation activity, you will receive 100%; 20 lectures - 100%, 19 lectures - $100 * \frac{19}{22} = 86\%$. Note: if you have an unpleasant communicable disease, please do not share!

Labs

You must be present in lab to complete the lab assignment. There is no way to make up for missing a lab. You will submit your lab assignments by pushing your work to your GitHub repository for the lab and submitting the PDF output on Gradescope *by the end of your lab session.*

The lowest two lab grades will be dropped at the end of the semester, which means you can miss up to two lab sessions with no penalty.

Exams

This course will have three in-class exams. You can demonstrate what you've learned in the course thus far through these exams. The exams will focus on both conceptual understanding of the content and application. The exam's content will be related to the content in lectures and labs.

Missed exams: There are no make-up exams. If you miss an exam due to an illness or similar issue documented with a Dean's Excuse, your final exam score will replace the missed exam score. If you miss the Final Exam due to an illness or similar issue documented with a Dean's Excuse, you will receive an Incomplete and can take the final exam later.

Improvement bonus: For students who take all exams (Exam 1, Exam 2, Exam 3, and the Final Exam), the final exam score will replace the lowest of the three mid-semester exam scores, if the final exam score is higher.

Assessment & Grading

Component	Weight	Details
Lectures	5%	Attendance/participation (20 minimum for full score)
Labs	20%	Individual or team-based, completed in class (lowest 2 dropped)
Exam 1	25%	In-class
Exam 2	25%	In-class
Exam 3	25%	In-class
Final	Replace lowest exam if desired	In-class

The final letter grade will be determined based on the following thresholds.

Letter Grade	Final Course Grade
A	[93,100]
A-	[90,93)
B+	[87,90)
B	[83,87)
B-	[80,83)
C+	[77,80)
C	[73,77)
C-	[70,73)

Letter Grade	Final Course Grade
D+	[67,70)
D	[63,67)
D-	[60,63)
F	[0,60)

Course policies

Duke Community Standard

As a student in this course, you have agreed to uphold the Duke Community Standard and the practices specific to this course.

Academic honesty

TL;DR: Don't cheat!

What is allowed and what is not?

Please abide by the following as you work on assignments in this course:

- **Collaboration:** Only work that is clearly assigned as teamwork should be completed collaboratively. On individual assignments, you may not directly share work (including code) with another student in this class; on team assignments, you may not directly share work (including code) with another team. "Sharing" includes, but is not limited to, messaging, emailing, or otherwise providing your work to another student or team.
  - Labs: Collaboration in teams for team lab assignments is not only allowed but expected. You will work together with your lab team to complete the lab exercise. However, each student must submit their own write-up of the lab exercise, which should reflect their understanding and ideas. It's expected that lab submission will be similar across team members.
  - Exams: You may not discuss or otherwise work with others on the exams while in progress. On exams, collaboration or using unauthorized materials will be considered a violation for all students involved, regardless of who initiated the sharing.
- **Use of online resources:** I am well aware that a huge volume of code is available on the web to solve any number of problems. Unless I explicitly tell you not to use something, the course's policy is that you may make use of any online resources, but you must explicitly cite where you obtained any code you directly use (or use as inspiration). **Any**

recycled code that is discovered and is not explicitly cited will be treated as plagiarism, resulting in an automatic 0 for the relevant portion of the assignment.

- **Use of generative artificial intelligence (AI)**: You should treat generative AI, such as ChatGPT, like other online resources. Two guiding principles govern how to use AI in this course:

  1. *Cognitive dimension:* Working with AI should not reduce your thinking ability.
  2. *Ethical dimension*: Students using AI should be transparent about their use and ensure it aligns with academic integrity.

  – ✔ **AI tools for code:** You may use generative AI tools when you need help with assignments. However you should first attempt to solve the problem yourself. Your submission should not be a copy-paste of AI-generated content – you must edit the content to ensure it reflects your understanding, has your voice and intellectual input, and conforms with course materials, syntax, terminology, and style. Additionally, **the prompt you use should not be copied and pasted directly from the assignment; you should create a prompt yourself.**

  It is also good practice to cite work submitted that is based on AI-generated content.

  Course materials are revised with assistance from OpenAI's ChatGPT (GPT-5.6 Sol) and Anthropic's Claude (Sonnet 5), accessed in summer and fall 2026.

While you're welcomed to ask AI tools questions that might help your learning and understanding in this course, you should be critical of the answers you receive, as AI-generated content may not always be accurate or reliable. Use it to supplement your understanding, not as a substitute for learning. You are ultimately responsible for the work you turn in; it should reflect *your* understanding of the course content.

What happens if you violate the academic honesty policy?

Any violations in academic honesty standards as outlined in the Duke Community Standard and those specific to this course

- will automatically result in a 0 for the relevant portion or the entire assignment or assessment,
- can result in further deductions to your overall course grade (e.g., drop down to the next letter grade or drop down to an F), and
- can be reported to the Office of Student Conduct & Community Standards for further action.

Regardless of course delivery format, it is the responsibility of all students to understand and follow all Duke policies, including academic integrity (e.g., completing one's own work, following proper citation of sources, adhering to guidance around group work projects, and more). Ignoring these requirements is a violation of the Duke Community Standard. Any questions and/or concerns regarding academic integrity can be directed to the Office of Student Conduct and Community Standards at conduct@duke.edu.

Regrade requests

Regrade requests must be submitted on Gradescope within a week after an assignment is returned. Regrade requests will be considered if there was an error in the grade calculation or if a correct answer was mistakenly marked as incorrect. Requests to dispute the number of points deducted for an incorrect response will not be considered. Regrade requests are also not a mechanism for asking for clarification on feedback, those questions should be brought to office hours. Note that by submitting a regrade request, the **entire assignment may be regraded**, which could potentially result in losing points.

No grades will be changed after the final exam has been administered.

Accommodations

Academic accommodations

If you need accommodations for this class, you will need to register with the Student Disability Access Office (SDAO) and provide them with documentation related to your needs. SDAO will work with you to determine what accommodations are appropriate for your situation. Please note that accommodations are not retroactive and disability accommodations cannot be provided until a Faculty Accommodation Letter has been given to me. Please contact SDAO for more information: sdao@duke.edu or access.duke.edu. Please note that the testing center can fill, so you want to go ahead and schedule accommodations on exam days (overlapping with our exam time) as soon as possible.

Religious accommodations

Students are permitted by university policy to be absent from class to observe a religious holiday. Accordingly, Trinity College of Arts & Sciences and the Pratt School of Engineering have established procedures to be followed by students for notifying their instructors of an absence necessitated by the observance of a religious holiday. Please submit requests for religious accommodations at the beginning of the semester so that we can work to make suitable arrangements well ahead of time. You can find the policy and relevant notification form here: trinity.duke.edu/undergraduate/academic-policies/religious-holidays